The empirical recurrence for OEIS A305243, recovered from its tail

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Abstract

OEIS A305243 records "Empirical recurrence of order 98" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 327$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 105$. Its last coefficient is $c_{98} = 8 \neq 0$, so no further tail satisfies anything shorter; any order-98 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A305243 is "Number of nX6 0..1 arrays with every element unequal to 0, 1, 2, 4 or 8 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$32, 184, 158, 767, 3288, 12521, 55039, 238455, \ \ldots$$

The entry states:

> Empirical recurrence of order 98 (see link above)

As of the "Last modified" line on the live entry (May 28 2018 08:41:58, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \; = \; \iota^{\top} M^{\, j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 327$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 327$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 98: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 98.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 654$ exact terms from $n = 7$ on, it returns order 98 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{98} c_i\, a(n-i),$$

c_1	2	c_{26}	160488	c_{51}	2073671	c_{76}	−71835
c_2	8	c_{27}	181038	c_{52}	−1657804	c_{77}	−69990
c_3	17	c_{28}	−173754	c_{53}	1859915	c_{78}	−40854
c_4	23	c_{29}	346083	c_{54}	−3231111	c_{79}	−83879
c_5	−143	c_{30}	−688375	c_{55}	−1685589	c_{80}	−82766
c_6	−326	c_{31}	−273323	c_{56}	119366	c_{81}	−14891
c_7	−502	c_{32}	416519	c_{57}	−5292536	c_{82}	−31965
c_8	−57	c_{33}	−597217	c_{58}	1011893	c_{83}	−17236
c_9	2720	c_{34}	1161923	c_{59}	−1574949	c_{84}	8954
c_{10}	3895	c_{35}	417357	c_{60}	−1719322	c_{85}	11497
c_{11}	5510	c_{36}	511160	c_{61}	2634371	c_{86}	10337
c_{12}	613	c_{37}	274023	c_{62}	−2170522	c_{87}	4934
c_{13}	−16568	c_{38}	−242776	c_{63}	2093766	c_{88}	9495
c_{14}	−19252	c_{39}	624312	c_{64}	1159912	c_{89}	1429
c_{15}	−40444	c_{40}	−2373362	c_{65}	−1298077	c_{90}	303
c_{16}	−19351	c_{41}	359981	c_{66}	2417657	c_{91}	536
c_{17}	49140	c_{42}	−1214489	c_{67}	−364797	c_{92}	−1237
c_{18}	48636	c_{43}	−2148050	c_{68}	454121	c_{93}	150
c_{19}	141205	c_{44}	598844	c_{69}	605120	c_{94}	−349
c_{20}	87980	c_{45}	−543458	c_{70}	−299986	c_{95}	−111
c_{21}	−53008	c_{46}	2007426	c_{71}	696471	c_{96}	66
c_{22}	−51818	c_{47}	400538	c_{72}	−341172	c_{97}	0
c_{23}	−221212	c_{48}	2885847	c_{73}	−41640	c_{98}	8
c_{24}	−118225	c_{49}	1760180	c_{74}	157838		
c_{25}	−80765	c_{50}	1206273	c_{75}	−169265		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 105$, and it is the recurrence OEIS A305243 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{98} - c_1 x^{98-1} - \cdots - c_{98}$. Its constant term is $-c_{98} = -8$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 98 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 98, so $\deg q = 98$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 98, so $q = q_{\min}$. ☐

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 98, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 8.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-98 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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